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Key Points:

- Snows in the upper Great Lakes have differing microphysical characteristics depending on their formation mechanism
- Synoptic states can be grouped together, and composite snow microphysical characteristics for each state can be determined
- Snow microphysics impact NEXRAD radar returns, which affects reflectivity to snow water liquid equivalent (*Z*-to-*S*) relationships

Correspondence to:
T. J. Wagner,
tim.wagner@ssec.wisc.edu

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The Synoptic, Microphysical, and Radar Characteristics of Upper Great Lakes Snow Events

Timothy J. Wagner[1], Jack Richter[2], Marian Mateling[3], Claire Pettersen[2], and Mark Kulie[4]

[1]Cooperative Institute for Meteorological Satellite Studies (CIMSS), Space Science and Engineering Center, University of Wisconsin–Madison, Madison, WI, USA, [2]Department of Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA, [3]Department of Atmospheric and Oceanic Sciences, University of Wisconsin–Madison, Madison, WI, USA, [4]NOAA/National Environmental Satellite, Data, and Information Service (NESDIS)/Center for Satellite Applications and Research, Madison, WI, USA

Abstract The upper Great Lakes region is well-known for significant snowfall during the winter months. The microphysical characteristics of the snow depend to a great degree on snow particle formation and growth processes that are linked to regime-dependent environmental conditions. A self-organizing map (SOM) was generated from over a decade of mean sea level pressure reanalyzes during winter months to classify the environment into lake-effect, synoptically forced, or ambiguous regimes. These regimes were then evaluated with respect to snow microphysical and bulk accumulation observations at Marquette, Michigan, near the southern shore of Lake Superior, and linked to radar-derived quantitative precipitation estimation (QPE) variability. Once event categories are defined by the SOM, it is possible to evaluate the microphysical characteristics of the different regime types. Synoptic snow forms over a deeper layer, has higher radar reflectivities, and features different particle size distributions when compared to lake-effect snow events. The disparate microphysical characteristics mean that the different regimes exhibit distinct radar reflectivity to snow liquid water equivalent rate (*Z*-to-*S*) relationships. Existing *Z*-to-*S* relationships used for operational QPE do not adequately represent both synoptic and lake effect snow, as they tend to overestimate snow from lake effect events and underestimate synoptic snowfall at higher reflectivity values. It is recommended that operational radar users adjust their *Z*-to-*S* relationship to match the regime-dependent forcing mechanisms.

Plain Language Summary Both lake-effect and synoptically forced snow is common in the upper Great Lakes region of the United States and Canada. However, the snow that forms during these two different event types can be quite different microphysically, which causes these events to appear differently on weather radars. In this study, the relationship between event types, snow microphysics, and radar signature is explored so that meteorologists can better use their radars to determine snowfall intensity and event accumulations more accurately.

1. Introduction

The Great Lakes of North America are a vast interconnected system of freshwater lakes that collectively contain over one-fifth of the planet's surface fresh water by volume. The weather and climate of the surrounding region is strongly influenced by the lakes due to their large thermal mass, significant latent heat fluxes, orography, and other processes. These influences can range from mesoscale events like the lake breeze (e.g., Wagner et al., 2021) to modifications of synoptic scale systems (Angel & Isard, 1997; Sousounis & Fritsch, 1994). With over 38 million people living in the drainage basin of the Great Lakes (Fergen et al., 2022) including over one-third of the population of Canada, and millions more living close enough to the lakes to experience lake-modified meteorological conditions, properly understanding the interactions between the lakes, the surrounding land, and the atmosphere is critical.

One of the most prominent meteorological features of the Great Lakes region, and one where the lakes exert a well-recognized influence, is cold-season snow events. Much of the snowfall in this region is produced by lake-effect snow (LeS): cold air outbreaks over comparatively warm open water surfaces can produce shallow but sometimes intense precipitation, often enhanced by orography (Agee & Hart, 1990; Eichenlaub, 1979; Geerts et al., 2022; Kristovich & Steve, 1993; Kristovich et al., 2017; Kulie et al., 2021; Mateling et al., 2023; Minder et al., 2015; Niziol et al., 1995). Individual LeS events can produce localized large amounts of accumulation

(Kulie et al., 2021; Niziol, 1987; Peace & Sykes, 1966), and downwind communities often receive more than 2 m of total snow accumulation annually (Braham & Dungey, 1984; Eichenlaub, 1979). The economic effects of these events can be significant (Kunkel et al., 2002; Schmidlin, 1993) as it can take many days for a community to recover from such intense snowfalls. While the underlying LeS formation mechanisms and associated large-scale synoptic conditions are generally understood (Hjelmfelt, 1990; Kristovich & Laird, 1998; Niziol et al., 1995), LeS remains a troublesome forecasting problem due to the variable mesoscale spatial structure of these events (Laird et al., 2003; Steiger et al., 2013). Accurate LeS event forecasts are further complicated by the inability of numerical weather prediction (NWP) models to consistently forecast the exact location, intensity, duration, and total snowfall production of the snow bands (Conrick et al., 2015; Reeves & Dawson, 2013). Furthermore, accurately forecasting snow-to-liquid ratios (SLR) for LeS is extremely difficult, as they can vary dramatically between events (Pettersen, Bliven, et al., 2020; Pettersen, Kulie, et al., 2020; Kulie et al., 2021).

Not all winter snowfall events in the Great Lakes region are forced by lake-air interactions, however. Many snow events are synoptically driven (Leathers & Ellis, 1996; Pettersen, Kulie, et al., 2020; Serreze et al., 1996), with snow formation and accumulation a result of processes that are commonly seen in wintertime midlatitude cyclones throughout the central and eastern United States. In general, these synoptically forced snowfall events are formed over greater vertical depths than the lake-effect snows (e.g., Kulie et al., 2021; Pettersen, Kulie, et al., 2020), and their impact can be experienced over far greater areas than the narrowly focused lake-effect bands found downstream of the Great Lakes. Additionally, snowfall production in synoptic events can be further enhanced by lake interactions and orographic uplift (e.g., Kulie et al., 2021; Owens et al., 2017). These lake-enhanced synoptic events are often associated with prolific snowfall production in the upper Great Lakes (Kulie et al., 2021). Due to the different growth processes (e.g., deposition, aggregation, riming, sub-cloud base sublimation) present for these snow event types, the microphysical characteristics of the LeS and synoptic snow events are distinct (Kulie et al., 2021).

Given the hazards that intense snowfall can present, it is critical that operational weather forecasters be able to monitor the extent, strength, and evolution of various types of snow events. One particularly useful tool for forecasters in the Great Lakes region is the Next Generation Weather Radar (NEXRAD) network operated by the National Weather Service (NWS) (Klazura & Imy, 1993). Their sensitivity to snowflake-sized particles and their rapid surveillance time is sufficient to monitor the evolution of small-scale lake effect. The reflectivity observations from the WSR-88D radars that comprise the NEXRAD network are important for qualitatively assessing snow event locations and intensity. Additionally, quantitative information about the snowfall rate can be derived through commonly used radar reflectivity (Z) to liquid water equivalent (LWE) snowfall rate (S) relationships (Ryzhkov & Zrnic, 2019), commonly called Z-to-S relationships, for quantitative precipitation estimation (QPE). A one-size-fits-all approach is ill-suited for Z-to-S relationships as the microphysical properties of snow vary substantially depending on snow event type (Cooper et al., 2017; Hiley et al., 2011; Huang et al., 2015; Kulie & Bennartz, 2009; Liu, 2008; Matrosov, 2007; Matrosov et al., 2009; von Lerber et al., 2017). It is these snow microphysical characteristics that determine the bulk scattering properties controlling radar signatures associated with surface snowfall events (e.g., Kneifel et al., 2011, 2015; Kulie et al., 2010, 2014; Kuo et al., 2016; Leinonen et al., 2012; Liu, 2008; Matrosov et al., 2019; Olson et al., 2016). Despite this known variability, snowfall rate estimates from state-of-the art network-based datasets like the Multi Radar Multi Sensor (MRMS; Zhang et al., 2016) currently adopt a static Z-to-S relationship:

$$Z = 75\, S^{2.0} \tag{1}$$

for all snowfall events regardless of type, where Z is the radar reflectivity factor in $mm^6\ m^{-3}$ and S is the LWE snowfall rate in $mm\ hr^{-1}$. NWS offices can, and do, apply regional Z-to-S relationships. For instance, multiple NWS forecast offices employ a Z-to-S relation tuned for the Great Lakes region:

$$Z = 180\, S^{2.0} \tag{2}$$

that is markedly different from the MRMS. The wide range of Z-to-S relations reflects the aforementioned microphysical variability among different snow events. In fact, recent work has shown the dependence of these parameters on the precipitation microphysics, with Bukovčić et al. (2018, 2020) specifically noting a dependence of the multiplicative constant in Equations 1 and 2 on the total ice concentration and particle size distribution.

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The present work empirically investigates various snow-producing weather regimes in the upper Great Lakes from both synoptic and microphysical perspectives. A self-organizing map (SOM) is created to help categorize ambient synoptic conditions into one of several similar states, or nodes. From that, the long-term deployment of a suite of snow microphysical instruments at the NWS Weather Forecast Office in Marquette, Michigan (MQT, Kulie et al., 2021; Pettersen, Kulie, et al., 2020) is used to evaluate key microphysical characteristics of the snowfall formed during the different SOM nodes. These instruments are located at the NWS Weather Forecast Office (WFO) in Marquette, Michigan, which facilitates an investigation of the validity of extant Z-to-S relationships for the various snow regimes and to determine which Z-to-S relationships are appropriate for a given event type. Section 2 discusses the instruments used in this study, while Section 3 discusses the SOM and interprets the ensuing classifications. Section 4 describes the microphysical characteristics of each SOM node, and Section 5 investigates the Z-to-S relationships for the various large-scale regimes determined by the nodes. A summary of this work and outlook for its application in operational milieus is contained in Section 6.

2. Instrumentation

This work utilizes observations from a continuous, multi-year suite of instruments at the Marquette, Michigan MQT NWS WFO, 13 km inland of Lake Superior and 426 m above sea level (Kulie et al., 2021). A full description of the Marquette snow microphysics observatory is given in Pettersen, Kulie, et al. (2020) and Pettersen, Bliven, et al. (2020). Brief descriptions of the most relevant instruments, the Precipitation Imaging Package (PIP) and Micro Rain Radar (MRR) are given here to aid in the interpretation of the results shown in this paper.

2.1. The Precipitation Imaging Package (PIP)

Operationally in the United States, LWE snowfall rates are typically measured by heated tipping bucket rain gauges or weighing bucket gauges. However, these systems typically have slow response times (Boudala et al., 2017; Kochendorfer et al., 2020) while wind often prevents snow from entering the gauge (e.g., Fortin et al., 2008; Goodison, 1978; Kochendorfer et al., 2021; Kochendorfer, Nitu, et al., 2017; Kochendorfer, Rassmussen, et al., 2017) or scours it out of the gauge before it has the chance to melt (Savina et al., 2011). Due to the difficulty of measuring instantaneous LWE snowfall rates, it can be challenging to empirically derive and validate Z-to-S relationships.

By contrast, novel instrumentation developed by the National Aeronautics and Space Administration (NASA) enables better observations of snowfall microphysics and LWE. The NASA Precipitating Imaging Package (PIP, Newman et al., 2009; Pettersen et al., 2021; Pettersen, Bliven, et al., 2020; King et al., 2024) is a ground-based video disdrometer. The PIP uses a coupled bright light source and high-speed camera to capture videos of falling hydrometeors. Image processing of these videos produces tables of hydrometeor microphysical characteristics at 1-min resolution, including particle size distributions (PSDs) and particle velocity distributions (VVDs) (Pettersen, Bliven, et al., 2020). The PIP can resolve hydrometeors ranging from 0.4 to 26 mm with a 0.2 mm resolution. Additional processing provides an effective particle density, which is retrieved from observing the particle size, shape, and fall speed. LWE precipitation rates are also able to be quantified (Pettersen, Bliven, et al., 2020), and PIP can effectively discriminate between precipitation phase (Pettersen et al., 2021). The enhanced observational capabilities of the PIP enable more finely resolved measurement of the LWE snowfall rate over traditional methods. By collocating PIP instruments with NEXRAD and profiling radars, it is possible to develop robust Z-to-S relationships for a variety of snow event types.

2.2. The MicroRain Radar (MRR)

The METEK MicroRain Radar 2 (MRR) is a 24 GHz (K band) vertically profiling frequency-modulated, continuous wave Doppler radar (Klugmann et al., 1996). The MRR is portable. relatively inexpensive, and uses relatively low power, thus making it useful for observing precipitation across remote regions. The MRR obtains profile observations of reflectivity, Doppler velocity, and Doppler spectral width at 1-min resolution (Klugmann et al., 1996). The operating range height of the MRR deployed at MQT is 3 km above ground level (AGL) and the resolution is 100 m. Observations below 400 m AGL are removed due to ground clutter contamination. Additionally, the MRR observations are processed using the Maahn and Kollias (2012) algorithm to enhance the detection of light snow and precipitation. The MRR is collocated within 10 m of the PIP instrument. Previous

Table 1
Details of the Self-Organizing Map Used in This Analysis

Node arrangement	3 × 3
Latitude limits (°N)	35 to 57
Longitude limits (°E)	−106 to −70
Performance function	Mean squared error
Epochs	200
Training function	Batch unsupervised

work used observations of precipitation echo-top height from the MRR to differentiate between synoptically forced (deep) and surface-driven (shallow) snow regimes (Mateling et al., 2021; Pettersen, Kulie, et al., 2020) as well as the influence of atmospheric rivers on snow development and accumulation (Richter et al., 2025). Additionally, the stability of the MQT MRR calibration over a 10-year period showed no drifting when compared with the MQT WSR-88D (Shates et al., 2023).

3. Event Classification and Identification

While the specific meteorological conditions of individual events are unique, many snow events share similar large-scale forcings. To help discern characteristics of different snowfall event types, it is useful to group events together based on common characteristics. While one could take a subjective approach to this process by inspecting individual meteorological charts and manually sorting them into discrete groups, this method is both tedious and subject to errors and biases. Other methods, such as principal component analysis (e.g., Cannon et al., 2002) or the Spatial Synoptic Classification (SSC) system (Sheridan, 2002) have been developed to reduce manual labor and provide a more reproduceable approach for synoptic regime classification. It is important to note that no method of synoptic classification is truly objective, as decisions still need to be made about the inputs into the classification scheme and how they are implemented.

In this paper, a self-organizing map (SOM, Kohonen, 1990) approach was employed to group similar patterns together. A SOM is an unsupervised machine learning technique that enables the sorting of multidimensional data into different categories without needing to define the characteristics of those categories in advance. There is a long history of SOM use in synoptic classification (e.g., Gallagher et al., 2020; Hewitson & Crane, 2002; Sheridan & Lee, 2011) as SOMs provide a convenient technique for reducing the continuous nature of ever-evolving atmospheric states into discrete categories with a minimum of human interference. They are also relatively simple to execute compared to other machine learning algorithms.

Mean sea level pressure (MLSP) reanalysis maps (Copernicus Climate Change Service, 2023) from the fifth generation of the European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis (ERA5; Hersbach et al., 2020) were used as inputs into a two stage SOM with a final grid-spacing of 3 × 3. The input data consisted of the months of November through February from 2014 to 2023. Only pressure maps representing times of active snowfall were used. While the ERA5 reanalysis provides data on an hourly scale, the aforementioned challenges associated with automated snowfall measurement meant that only manual observations of snow (as carried out through manual measurements of snow accumulation at the NWS MQT office) were preferred as an independent indictor the occurrence of snow during the representative time of a particular input map. These observations are carried out every six hours; therefore, the final SOM only featured inputs from 05:00, 11:00, 17:00, and 23:00 UTC to correspond with these observation times, and at least 0.2 in of snow had to have been recorded for that time to be included in the analysis. Snowfall observations are used in lieu of estimates from ERA5 due to known discrepancies between reanalysis and observed snow accumulation. As the snow accumulation dataset is recorded with a precision of tenths of inches, this study uses inches instead of centimeters when working with the snow accumulation dataset in order to avoid creating false levels of precision that arise when converting between the two systems of units. The SOM was developed in two phases: the first phase created a 12 × 12 SOM to provide the opportunity to capture large degrees of variability in the dataset, while the second phase narrowed that 12 × 12 grid down to the 3 × 3 grid seen throughout this work to further refine the larger-scale synoptic patterns. Full details for the SOM are shown in Table 1. In this paper, the term "event" is used to refer to one of these six-hour periods where snowfall has been recorded. A single continuous snow storm may thus span multiple events as they are defined herein and can transition from one type of forcing to another throughout a storm's lifespan. Regardless, this SOM technique provides a useful methodology for grouping similar synoptic states together.

Figure 1 depicts the mean MSLP state for each of the nine nodes identified by the SOM. Through manual inspection, some commonalities between the nodes have been identified. For Nodes 4, 5, 6, 7, and 9 (outlined in red in Figure 1; 61.4% of all cases), significant flow over Lake Superior is present; it is therefore likely that snow events in these nodes are LeS in nature. In nodes 1 and 2 (outlined in blue in Figure 1; 17.2% of all cases), the composites indicate moderate-to-weak low pressure systems found in the Great Lakes basin. Flow at Marquette

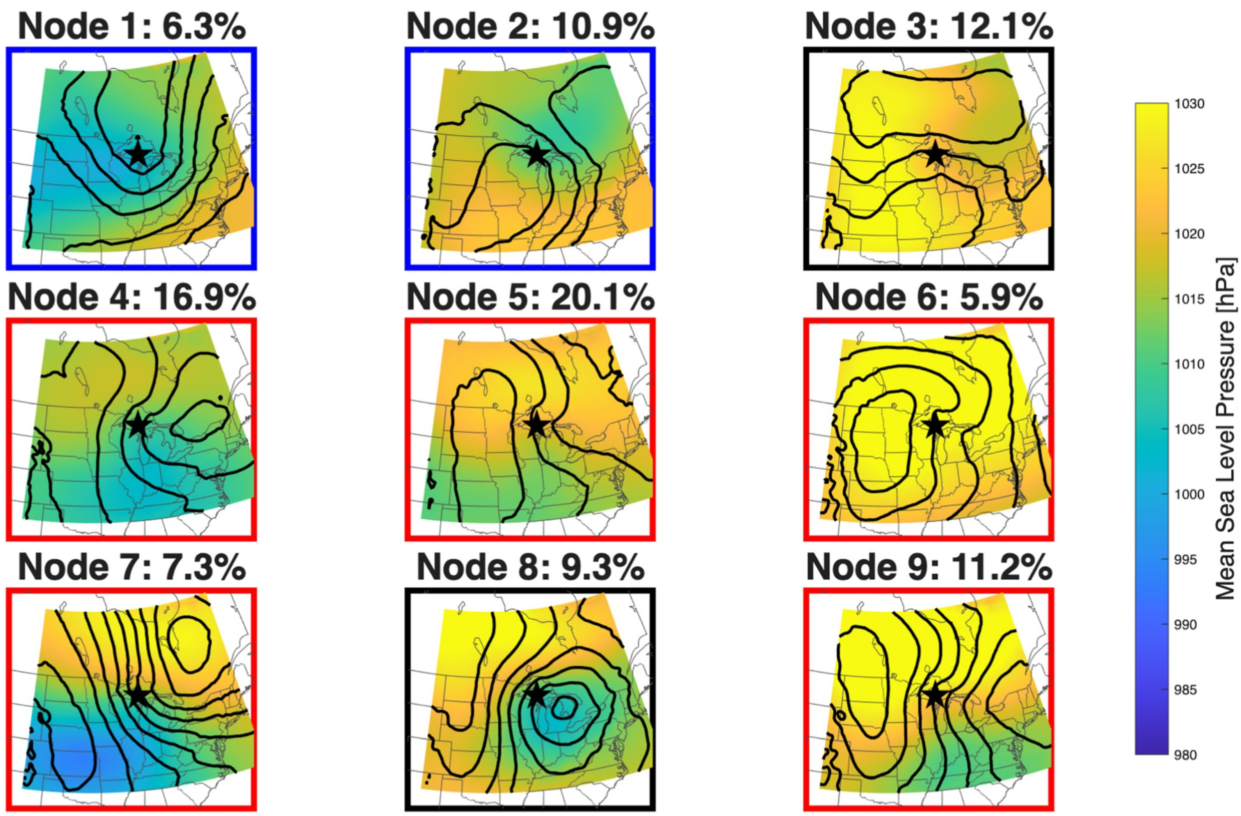

Figure 1. Mean MSLP maps (in hPa) for each of the nine nodes identified by the self-organizing map described in Section 3. The relative frequency of occurrence of each node is noted as a percentage above each map. The mean snowfall as observed at the Marquette, Michigan National Weather Service forecast office (KMQT) is indicated in each panel. The location of the Marquette forecast office is marked with a black star. Colored boxes are used to further identify node groups that were manually determined: flow over Lake Superior that is consistent with lake effect events (red), midlatitude cyclones without significant flow over Lake Superior (blue), and other (black).

(the black star in Figure 1) during these conditions has little to no fetch over Lake Superior, and thus snow during these cases is much more likely to be synoptically forced than lake-induced. Nodes 3 and 8 (outlined in black in Figure 1; 21.4% of all cases) feature environments with little local forcing. However, these ambiguous nodes, especially Node 3, share some traits with snow events enhanced with atmospheric rivers (Mateling et al., 2021). This includes flow at Marquette from the west or southwest. For the rest of this paper, these three groups will be referred to LeS, synoptic, and ambiguous. It is important to note that while a SOM gives the appearance of an objective analysis, there are elements of subjectivity in the design of the SOM and individual cases may fit better in nodes other than the ones they were assigned by the algorithm. Regardless, this is a valuable tool for assessing the bulk characteristics of grouped cases.

In addition to using the MSLP data products, the ERA5 temperature and water vapor flux data are analyzed on a synoptic scale. Anomalies of temperature at 850 hPa from the climatological mean (here defined as same 2014–2023 period used to generate the SOM) are computed. For each node, the monthly mean temperature at 850 hPa (T_{850}) is subtracted from each individual snow event's T_{850} to compute the anomaly, ΔT_{850}. The mean integrated vapor transport (IVT) for each node is computed by finding the resultant of eastward and northward water vapor fluxes from ERA5. Figures 2 and 3 show ΔT_{850} and mean IVT, respectively, for each node.

The ERA5 temperature anomalies at 850 hPa (ΔT_{850}, Figure 2) further contextualize the synoptic conditions for each node beyond what is seen in with the MSLP plots shown in Figure 1. Temperature at 850 hPa is less affected by diurnal variability compared to the surface temperature and is thus a better indicator of baroclinicity and warm and cold air masses than surface values (Mote, 2008; Suriano & Leathers, 2017). Figure 2 shows the mean ΔT_{850} for each node. For LeS events (red boxes; nodes 4, 5, 6, 7, and 9), Δ°C at Marquette is negative, indicating colder-than-average conditions. The larger magnitude of ΔT_{850} for nodes 6 and 9 are associated with a greater mean

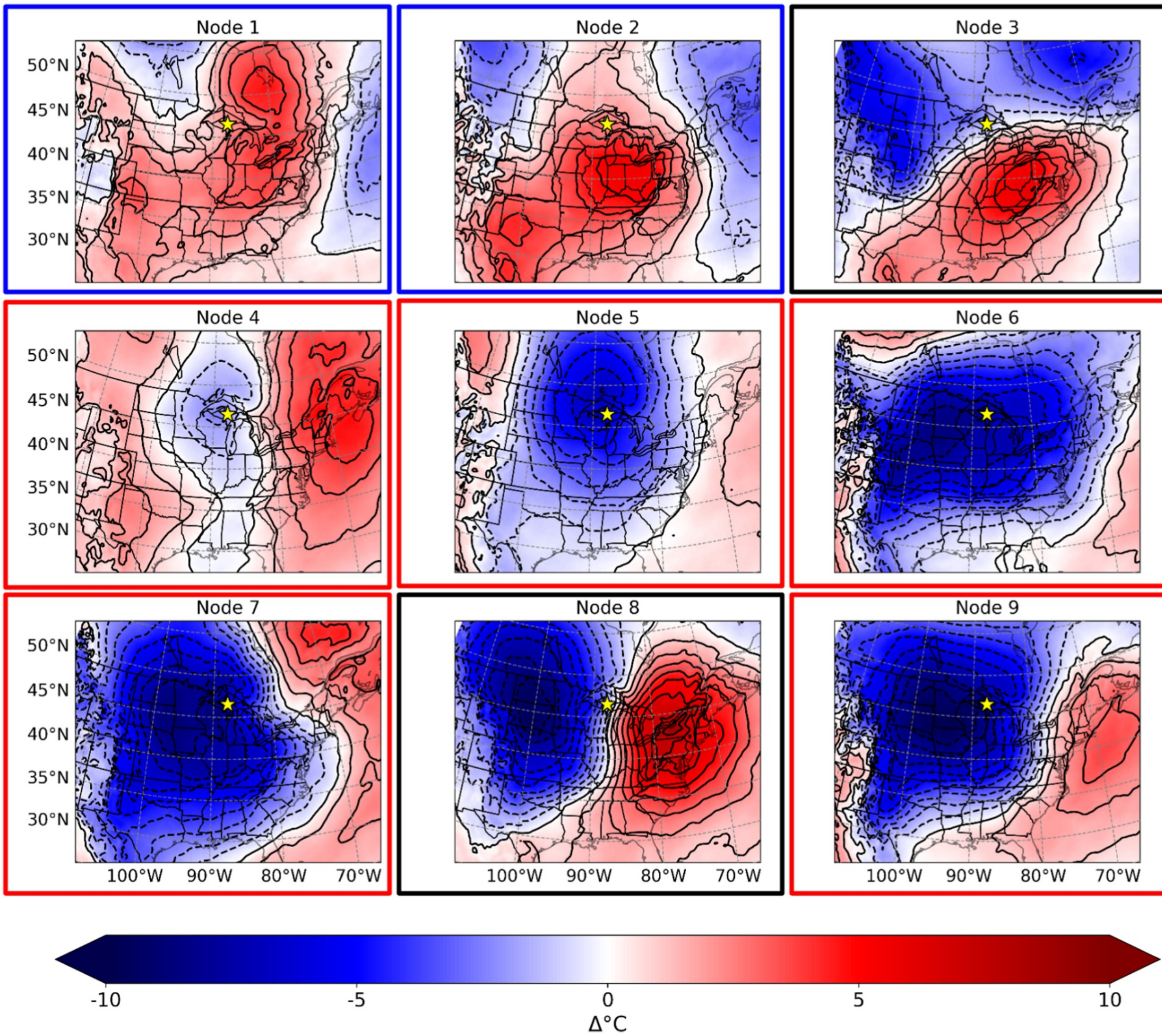

Figure 2. Composites of the climatological anomalies of 850 hPa temperature (ΔT_{850}) for each node, shown in Δ°C. The location of the Marquette forecast office is marked with a yellow star. The colored boxes around each node match the event categories defined in Figure 1.

MSLP to the west, whereas nodes 4 and 7 are associated with a lower mean MSLP to the east of Marquette (Figure 1). Node 5 shows larger negative magnitude of ΔT_{850}, though neither the high pressure to the west nor the low pressure to the east of Marquette is of large magnitude, relative to other LES nodes. Nodes 1 and 2 (blue boxes; synoptic) show anomalously warmer T_{850} in Marquette associated with low-pressure systems, but T_{850} for Node 1 is moderate low-pressure systems. While we have assigned Node 1 to the Synoptic category, its flow patterns are consistent with the lake-enhanced synoptic events investigated by Kulie et al. (2021) and Mateling et al. (2021). Events classified as nodes 3 and 8 (black boxes; ambiguous) have nearly 0 ΔT_{850}, but node 3 is associated with little forcing, whereas node 8 is associated with strong low-pressure systems (Figure 1).

To investigate where the moisture for these snowfall events originate, Figure 3 shows the mean integrated water vapor transport (IVT), which quantifies the magnitude and direction of the water vapor flux through an atmospheric column, in kg m^{-2} s^{-1}. The IVT amount at Marquette is notably different between nodes, but generally lower than the surrounding region. LeS nodes (4, 5, 6, 7, and 9; red boxes) have relatively low mean IVT values (greens) from the northwest, indicating that drier air advects across Lake Superior during these events in Marquette. Synoptic nodes (1 and 2; blue boxes) show larger magnitude mean IVT values (blues) from the southwest. Events classified as ambiguous (3 and 8; black boxes) show differing direction and magnitude of IVT at

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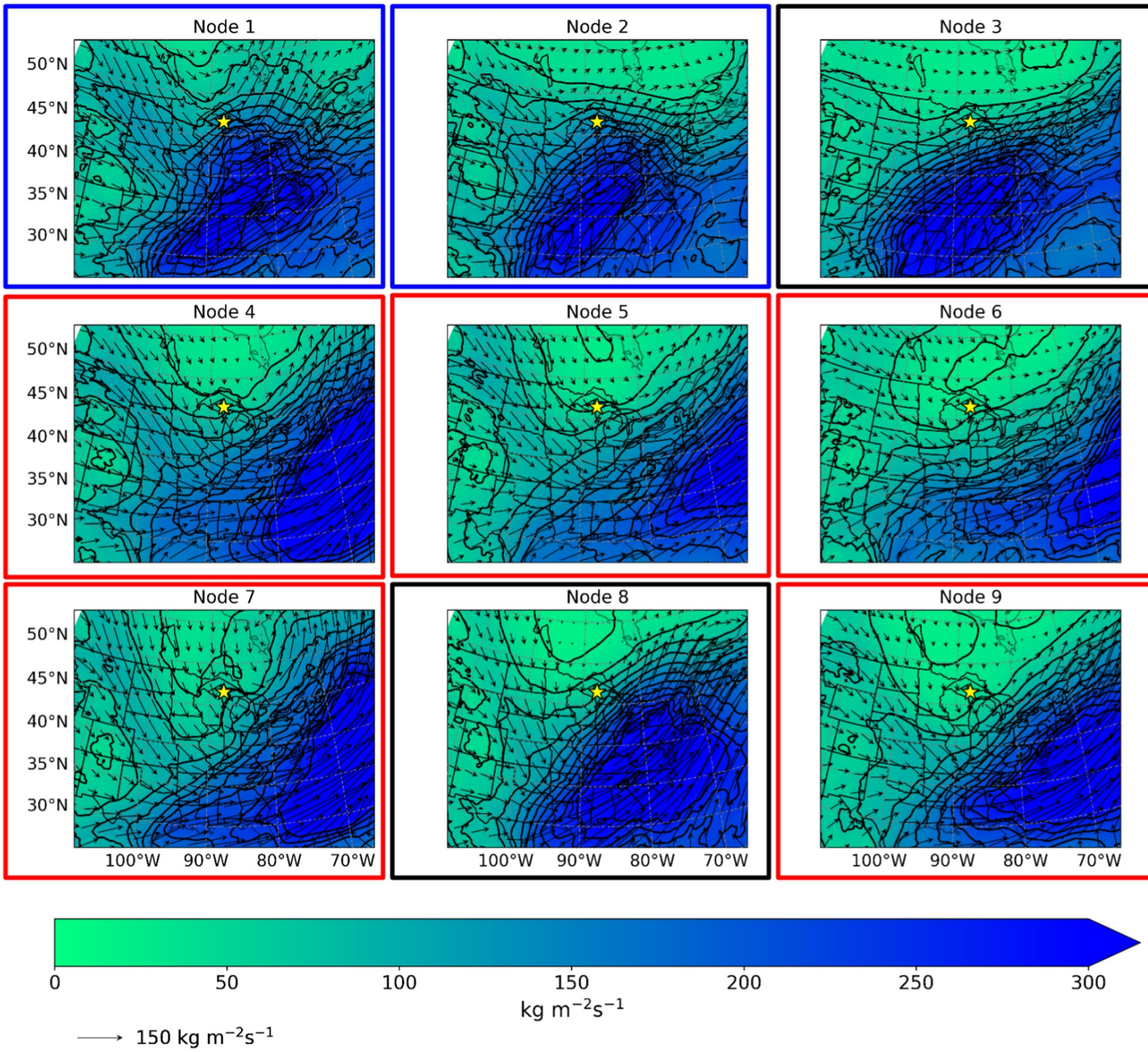

Figure 3. As in Figure 2, but for mean integrated water vapor transport (IVT).

Marquette. Node 3 shows large IVT from the southwest and moderate IVT from the Gulf of Mexico and the west, similar to synoptic nodes. Node 8 shows moderate IVT from the northwest, similar to LES nodes.

It may be instructive to observe the temporal distribution of events, as different nodes may be more common during different parts of winter. The frequency of occurrence in a given month for each node type is seen in Figure 4. As noted earlier, there are many more events in the lake effect (red) category than in the synoptic (blue) or ambiguous (black) classifications. Regardless of the lake effect node, there is a decrease in the number of lake effect snow events from January to February; while this is partially attributable to the fewer number of days in February a more salient factor is that the lakes tend to freeze later in the winter and thus exert less of an influence on local meteorological conditions due to a reduced gradient in air temperature between the maritime and continental air. By contrast, the ambiguous events are more likely to occur as winter continues. In total, 1,052 separate events are analyzed here.

4. Snow Regime Microphysical Variability

The suite of instruments at the Marquette site provides the ability to inspect the microphysical properties of the snow. By partitioning the microphysical characteristics by the previously defined SOM nodes, it is possible to

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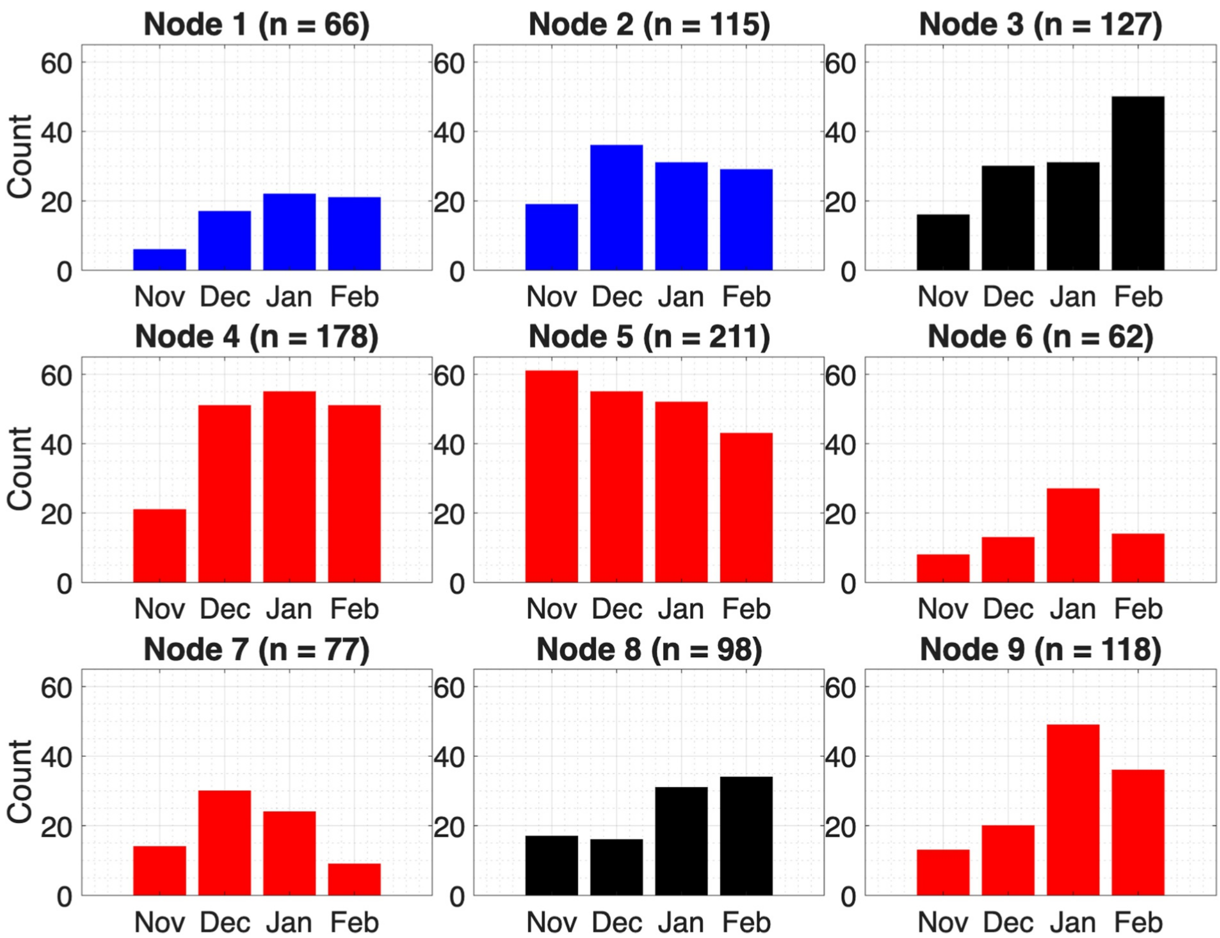

Figure 4. The distribution of months of occurrence for each of the node types shown in Figure 1 with the number of events in each node shown in parentheses. The colors of the bars are consistent with the event categories defined in Figure 1.

determine some of the snow particle characteristics associated with the different snowfall formation mechanisms. Figure 5 shows two-dimensional histograms of MRR reflectivity as a function of height for each of the node types for the years 2015 through 2022, corresponding to the years in which MRR (Richter & Pettersen, 2024) and PIP (King & Pettersen, 2023) were collocated, and the latter instrument was producing LWE snowfall rates. This plot enables understanding of the layer depth over which the snow is being formed.

Figure 5 demonstrates that the lake-effect events are forming over relatively shallow layers, with the bulk of radar returns originating in the lowest 1 km above ground level (AGL). The synoptic events stand in sharp contrast, with deep reflectivity returns (especially in the 5–10 dBz range) up to the 3 km limit of the MRR observations. Node 2 in particular shows frequent occurrence of high reflectivity values (10–20 dBZ) at all heights. From Figure 5, it is clear that snow particles are forming and growing over much deeper layers of the atmosphere for the synoptic snowfall events than the LeS ones, consistent with previous investigations of snowfall regimes (e.g., Pettersen, Kulie, et al., 2020; Shates et al., 2021). These histograms provide additional confidence in this study's choices of lake-effect, synoptic, or ambiguous for a given node. The ambiguous category (Nodes 3 and 8) merits special attention as it, like Nodes 1 and 2, has a large portion of its returns over a deep layer of the atmosphere. However, the reflectivity values associated with that deep layer of radar returns in the ambiguous nodes are higher than their synoptically forced counterparts, with high numbers of observations in the 10–20 dBZ range.

The PIP observations enable comparisons of the characteristic particle size distributions (PSDs) for each node. Figure 6 depicts the median, 25th, and 75th percentile PSDs for each of the node-aggregated composite PSDs. The 25th percentile and median PSDs show little variability between the nodes, and generally show low particle counts at all diameters with few particles above 2 mm for all nine nodes. These 25th percentile PSDs also have a

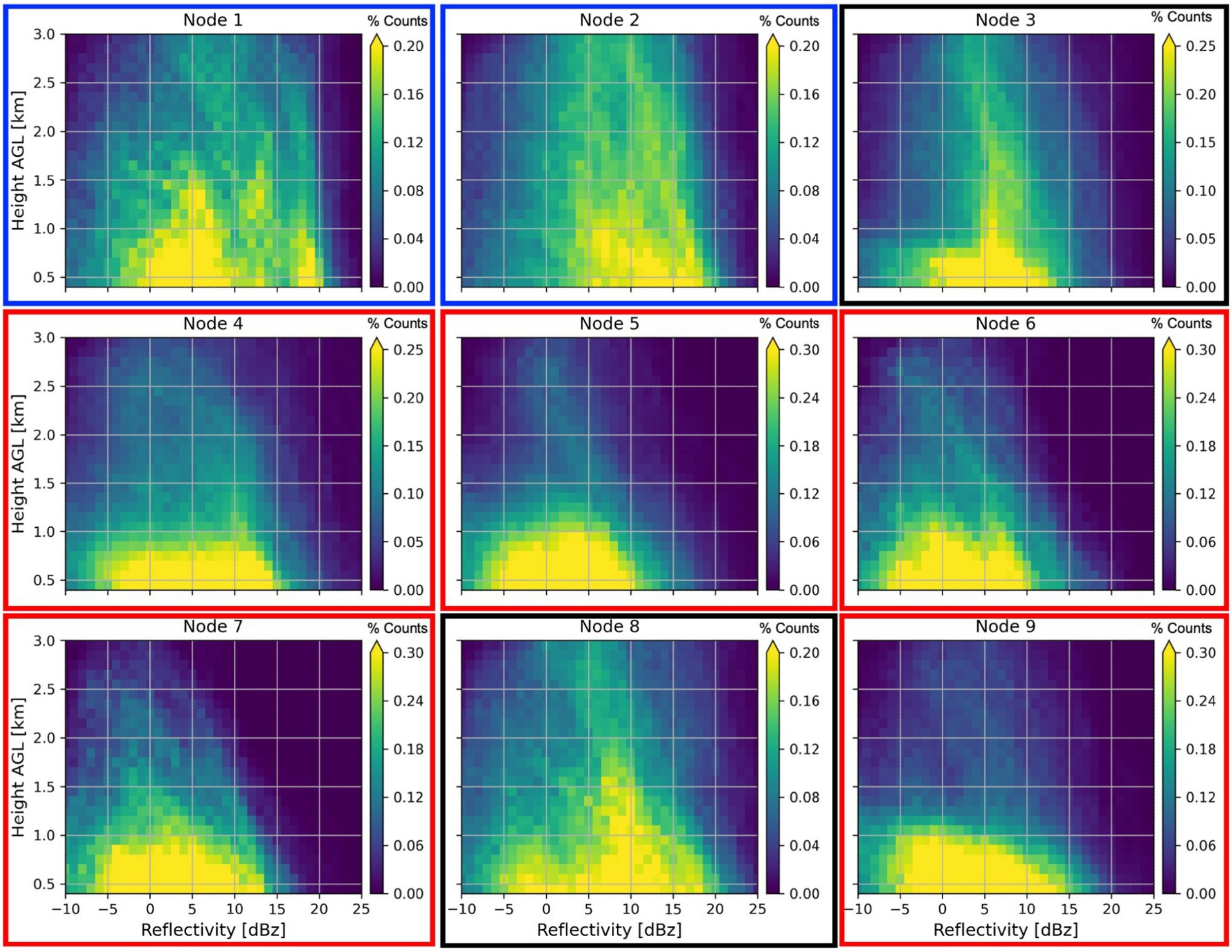

Figure 5. Two-dimensional histograms of radar reflectivity as a function of height above ground level for each of the nine nodes as observed by the MRR at the Marquette instrument site. To better enable qualitative comparison of the distribution shapes, the color scales represent raw counts and are unique to each panel. The colored boxes around each node are consistent with the event categories defined in Figure 1.

peak in the PSD around 0.6 mm particle diameter with peak number concentrations between 10^{2} and $10^{2.5}$ m^{-3} mm^{-1}. The median PSDs show few particles above 3 mm and have peak number concentrations between $10^{2.5}$ and $10^{3.0}$ m^{-3} mm^{-1}. More variability is seen in the characteristic PSDs at the 75th percentile between the nine nodes. Nodes 1, 2, 3, 4, and 9 have PSDs that are relatively broad at the 75th percentile whereas the others are relatively narrow at that level. The nodes with broad PSDs at the 75th percentile reveal the presence of particles >4 mm whereas the nodes with narrow PSDs at the 75th percentile instead show more particles at smaller diameters. While this demonstrates that disparate microphysical properties exist between the nodes, the 75th percentile PSDs do not appear to solely depend on the snowfall regime. The LeS nodes show the presence of both broad and narrow PSDs at the 75th percentile, which suggests that other factors such as the environmental conditions also influence the microphysical properties of these nodes.

To further explore PSD variability within and between each of the regimes, the median PIP PSDs of each regime stratified by PIP LWE snow rate and MRR near-surface reflectivity are examined (Figure 7). The stratification categories are defined to capture the PSD characteristics of light (0.0–0.5 mm hr^{-1}; <10 dBZ), moderate (0.5–1.0 mm hr^{-1}; 10–15 dBZ), and heavy (>1.0 mm hr^{-1}; >15 dBZ) snowfall for the three regimes. The median PSDs within the light snowfall category are effectively uniform for each regime, with few particles overall and the majority of particles <2 mm. The median PIP PSDs in the light snowfall category also appear gamma-distributed, in that the PSDs do not monotonically decrease with diameter and rather have a N_{max} around 0.6 mm that decreases on both sides. The median PSDs within the moderate snowfall category are also relatively uniform, but have more variability between the regimes compared to the light snowfall category. Specifically, there is a subtle

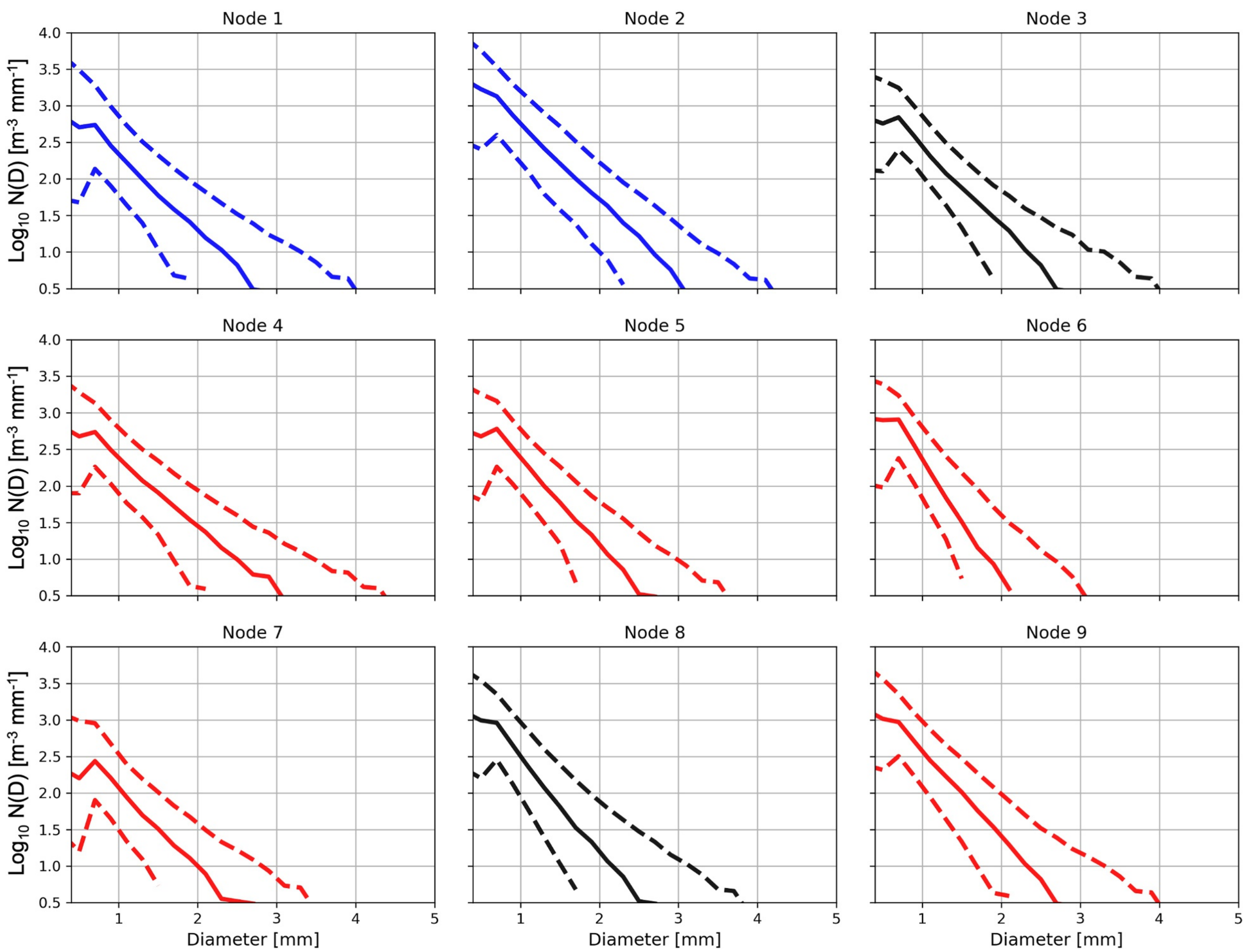

Figure 6. Characteristic particle size distributions (PSDs) for the nine nodes as observed by the PIP at the Marquette instrument site. On each node, the solid line represents the median PSD of the node whereas the dashed lines represent the 25th and 75th percentile PSDs of the node. The colors of the lines are consistent with Figure 1.

increase in the number of large particles for the LeS regime compared to the synoptic and ambiguous regimes, which is particularly visible in the moderate near-surface reflectivity stratification. Furthermore, the median PIP PSDs in the moderate snowfall category are inverse-exponential PSDs that monotonically decrease with increasing diameter. The median PIP PSDs in the heavy snowfall category have the most variability between the regimes. At small diameters, the LeS PSD has the least number of particles whereas the synoptic PSD has the greatest number of particles. In contrast, at large diameters, the LeS PSD has the greatest number of particles whereas the synoptic PSD has the least number of particles. The ambiguous regime PSD is located between the LeS and synoptic PSDs at all diameters.

These differences reveal the unique processes in which the three regimes produce heavy snowfall. The LeS regime induces heavy snowfall by producing more large particles whereas the synoptic regime induces heavy snowfall by producing more particles at small diameters and these differences are consistent with previous regime studies of snowfall (Pettersen, Kulie, et al., 2020; Shates et al., 2021). It is more difficult to characterize the process by which the ambiguous category induces heavy snowfall, but likely is some combination of the LeS and synoptic processes as its PSD straddles the two regimes.

5. *Z*-to-*S* Relationships as a Function of Snow Event Type

As noted above, *Z*-to-*S* relations can vary substantially based on the microphysical characteristics of the snow, which, in turn, are a function of large-scale forcing and the mechanisms controlling snow particle formation,

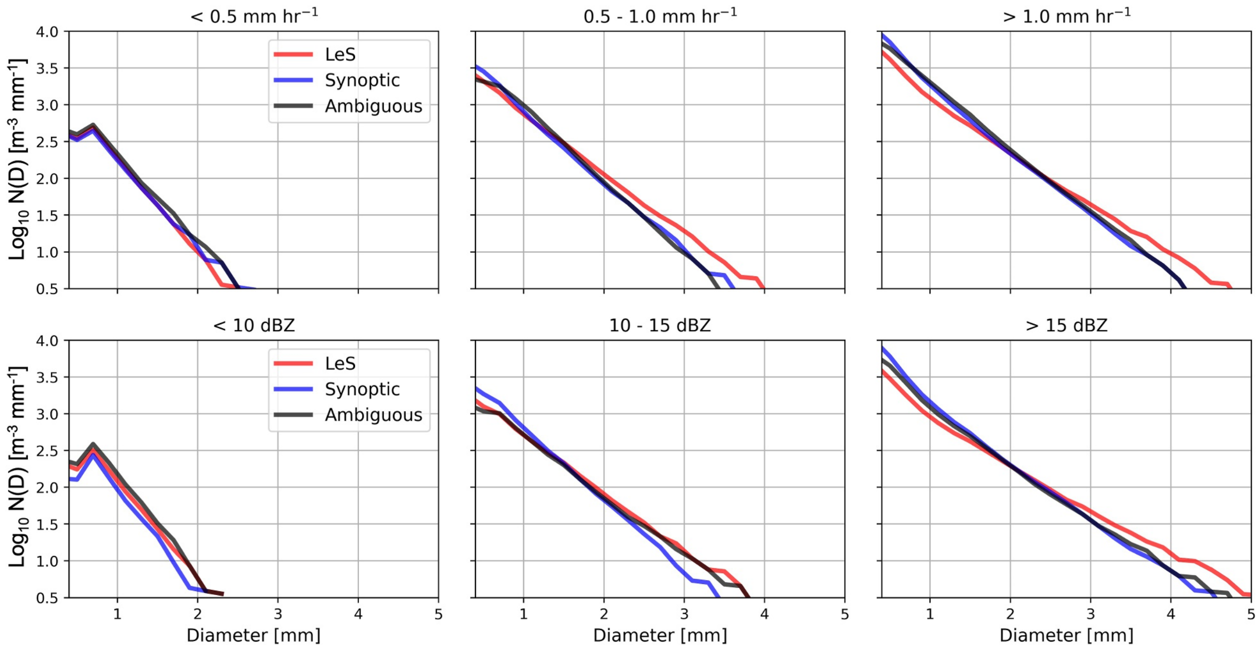

Figure 7. Median PIP PSDs of the three snowfall regimes stratified by PIP LWE snow rate and MRR near-surface reflectivity. The stratification categories are denoted at the top of each plot and the median PSD lines of the different snowfall regimes are colored consistent with previous figures.

growth, and evolution. Therefore, it is important to evaluate the Z-to-S relations across all SOM nodes to determine the impact that large-scale regime and snow event type manifest in the observed radar signatures. In order to do this, reflectivity values from the Marquette WSR-88D NEXRAD radar (NOAA National Weather Service Radar Operations Center, 1991) are matched to PIP-derived snowfall rates. Since the PIP and the WSR-88D are collocated at the Marquette forecast office, the PIP is contained within the cone of silence of the radar. Therefore, the mean reflectivity factor of the first valid ring in the 2.4 deg elevation scan is chosen as the representative reflectivity for the PIP site; this elevation angle was chosen as lower angles frequently had missing or masked values and is about 2.1 km azimuthally from the radar at a height of about 93 m above the radar. A 5 min running average of the PIP-retrieved LWE snowfall rate is computed to better match the temporal resolution of the radar as well as account for small-scale variability. For a given node, the radar and PIP-derived snowfall LWE rates are matched in time to create paired observations, which are then sorted into 1 dBZ reflectivity bins from 0 to 25 dBZ. The mean PIP LWE snow rate is calculated for each bin, and these values are used to determine a power law fit of the same form as Equations 1 and 2:

$$Z = a\,S^{b} \tag{3}$$

where Z is the reflectivity favor in mm^6 m^{-3}, S is the snowfall rate in mm hr^{-1}, and a and b are the coefficients of the fit. This process is then repeated for every node. Results of this analysis are shown in Figure 8.

It is clear from Figure 8 that the Z-to-S relations vary significantly between different nodes, and that the Great Lakes Z-to-S relation (Equation 2) only agrees with a handful of the nodes, namely those assigned to the ambiguous category. Of the lake effect nodes, Equation 2 captures the lower snowfall rates and reflectivities, but generally overestimates the LWE snow rates for the higher reflectivity observations. By contrast, Nodes 1 and 2 (both part of the synoptically forced regime) are clearly underestimated by Equation 2. In general, when the nodes are grouped together into LeS, synoptic, and ambiguous snow events, the general shape of the fits are similar for all nodes in a group. In essence, a particular node represents the variability within the grouped regime.

Further insight can be gained by aggregating all of the values for the LeS, synoptic, and ambiguous SOM nodes into a single Z-S relationship for each snow event type. These results are depicted in Figure 9, along with the Great Lakes fit (Equation 2) as well as the MRMS relationship (Equation 1). Both prescribed Z-to-S relations overestimate the amount of snow from LeS events for nearly all reflectivities; it is only at the highest reflectivities that

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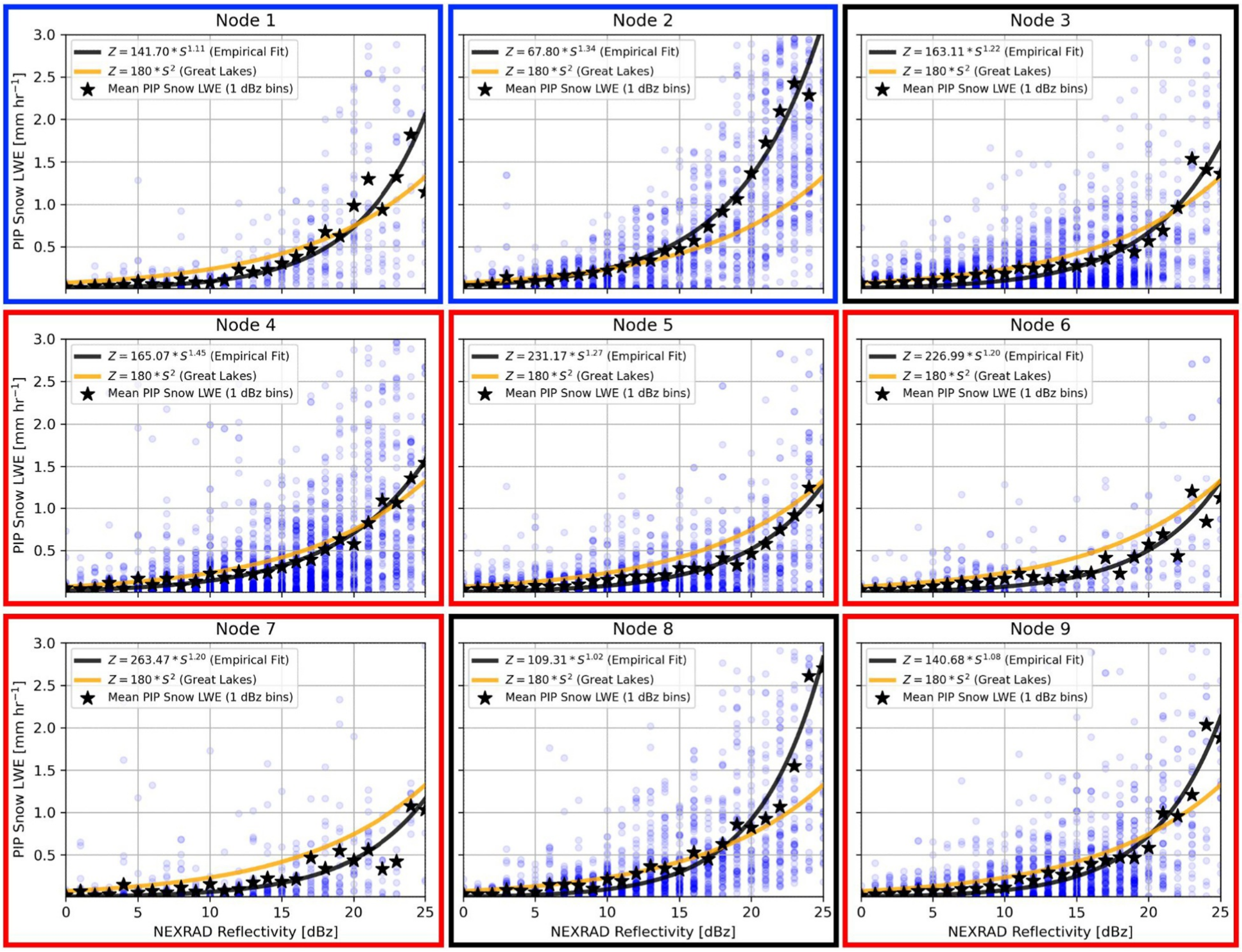

Figure 8. Empirically derived power law fits of snowfall rate to radar reflectivity (black) for each of the nine synoptic nodes. Fits are derived from the mean snowfall rate for each 1 dBZ bin (black stars). Individual *Z*-*S* pairs are shown as blue dots, while the Great Lakes fit from Equation 2 is shown in orange.

the Great Lakes relationship matches the observed values. By contrast, the Great Lakes relationship more effectively represents the synoptic snowfall rates for low-to-moderate reflectivity values, as Equation 2 and the empirical fit match very well up to ∼17 dBZ. Beyond that, however, the observed snowfall rates are much greater than what is predicted by Equation 2. At 20 dBZ and beyond, the MRMS does a better job of predicting snowfall rate than the Great Lakes relationship. Still, it is clear that forecasters can realize more accurate estimates of snowfall rate from radar if they use *Z*-to-*S* relationships that are tuned to specific conditions; if it is not clear what

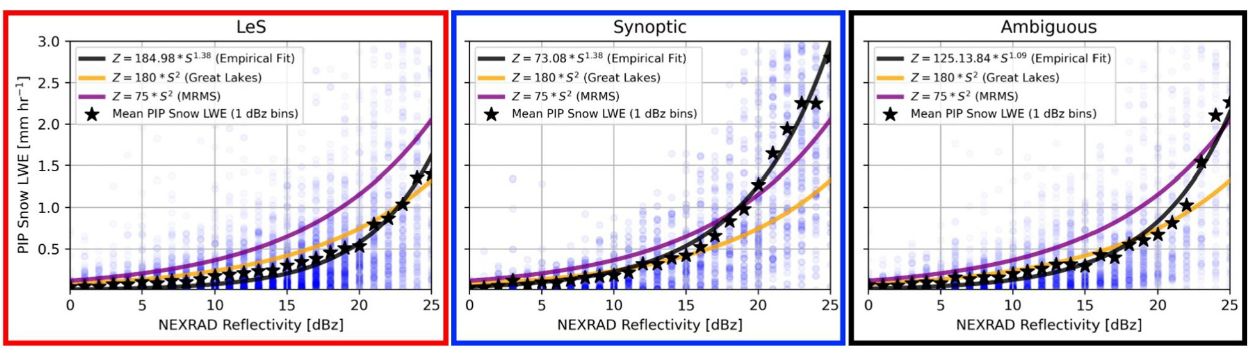

Figure 9. As in Figure 8, but for all lake effect (left), synoptic (middle), and ambiguous (right) events aggregated together. The MRMS *Z*/*S* relationship is also shown in purple.

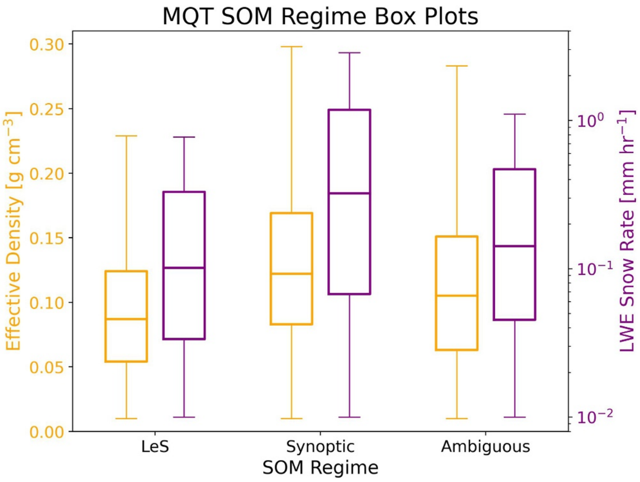

Figure 10. Box plots showing the distributions of effective density and snow rate for the three regimes. The orange boxes and features correspond to effective density whereas the purple boxes and features represent snow rate. The boxes extend from the first quartile to the third quartile, and the whiskers extend from the box to the farthest data point lying within 1.5x of the inter-quartile range on either side. The medians are denoted by the horizontal lines within the boxes. The effective density and snow rate data excludes values below 0.01.

regime is present in the current environment, then the Great Lakes relationship is most likely to produce accurate representations of the liquid water-equivalent snowfall rate over the widest range of reflectivities. It is important to note that individual nodes can and do show different behavior when compared to the aggregated values, in part due to differing sample sizes. Future work will investigate how small changes in synoptic-scale flow can have observable impacts on snow microphysics.

Box plots showing the distributions of effective density and snow rate for the three regimes are shown in Figure 10. The effective density measurements are calculated by the PIP, with higher values indicative of heavier and wetter snow and lower values indicative of lighter and fluffy snow. The synoptic and ambiguous regimes overall show higher effective density values compared to the LeS regime. The median effective density values for the synoptic and ambiguous regimes are about 0.12 g cm^{-3} whereas the median effective density value for the LeS regime is about 0.09 g cm^{-3}. This reveals that the synoptic and ambiguous regimes produce snowfall that is heavier and has a higher liquid water content than the LeS regime. Additionally, the synoptic regime has the most variability in effective density values compared to the other two regimes. 1.5x of the inter-quartile range (extent of the whiskers) for the synoptic regime extends from 0.01 to 0.29 g cm^{-3} while the whisker range extends from 0.01 to 0.26 g cm^{-3} for the LeS and ambiguous regimes.

The LWE snow rate measurements are also calculated by the PIP and reveal similar results as the effect density measurements. The synoptic and ambiguous regimes overall show higher LWE snow rates compared to the LeS regime. The median snow rate for the synoptic and ambiguous regimes is about 0.2 mm hr^{-1} and 0.15 mm hr^{-1} respectively while the median snow rate for the LeS regime is only about 0.1 mm hr^{-1}. Additional differences emerge between the three regimes at the higher percentiles and at the upper whiskers. The 75th percentile snow rates are about 0.3, 1.0, and 0.4 mm hr^{-1} for the LeS, synoptic, and ambiguous regimes, respectively. Moreover, the highest snow rates lying within 1.5x of the inter-quartile range are about 0.85, 2.0, and 1.0 mm hr^{-1} for the LeS, synoptic, and ambiguous regimes, respectively. These differences at the higher values indicate that the synoptic regime has the greatest tendency to produce high LWE snowfall rates and the LeS regime has the lowest tendency to produce high LWE snowfall rates.

6. Conclusions and Outlook

Snow remains a significant topic of interest to the Great Lakes region due to its impact on commerce, public safety, transportation, hydrology, and other domains. Depending on the mechanism that is forcing the snow in this area, the microphysics of the snow can be substantially different with noticeably distinct patterns in formation depth and particle size distributions, among parameters. These differences are a key component the disparate macrophysical characteristics of the snow regimes. Snow accumulations in lake effect events are lower than in synoptic ones, although the greater frequency of lake effect snows means that these events are responsible for more snowfall overall. The microphysical differences between different snow-forcing mechanisms presents a challenge for nowcasting emerging and ongoing snow situations.

A key effect of the distinct microphysical characteristics between snow regimes is the differing relationship between snowfall rate and observed radar reflectivity, commonly called the Z-to-S relationship. No single Z-to-S relationship is adequate for representing all snow rates at all reflectivities, and operational weather centers can achieve substantial improvement in snowfall estimates simply by selecting a Z-to-S relationship that has been tuned for the relevant snow regime, thus aiding their nowcasting efforts.

It is important to note that the Z-to-S relationships calculated here were calculated specifically at the Marquette, Michigan, site. Given local changes in terrain, water temperature, and other forcing, it is likely that SOMs calculated for other Great Lakes radars may appear slightly different. However, these are means to an end of automated identification of LeS and synoptically forced cases for further analysis. With the similarity in performance over a larger range of radar reflectivity values between the empirically derived Z-to-S relationship for synoptic snow presented here and the independently derived Great Lakes *Z*-to-*S* relationship, it is likely that the results presented here are largely transferable to other sites in the region.

One application for this work is the development of a machine learning approach for retrieving snowfall rate from satellite observations over the Great Lakes. Radar coverage over the lakes can be inconsistent: terrain near radars often contributes to beam blockage, the shallow nature of lake effect snow means it often forms below the lowest elevation angle from the radar, and the large size of the lakes means parts are beyond the reach of United States-based radars. A snow rate retrieval from geostationary satellites can ameliorate all of these issues. However, this requires an effective and accurate measure of snow rate to serve as truth in the training process. By using the MRMS gridded reflectivity product and applying the proper *Z*-to-*S* relationship depending on the active regime, a more accurate measure of the snow rate from radar can be obtained and a better satellite-based estimate can be determined. The research team is actively working on such a product at this time, and this analysis of *Z*-to-*S* relationships is a key component of that process.

Conflict of Interest

C. Pettersen is an associate editor of JGR-Atmospheres.

Availability Statement

All data used in this study were obtained from publicly accessible sources. The ERA5 reanalyses used to generate the SOM were downloaded from the Copernicus Climate Data Store (https://doi.org/10.24381/cds.adbb2d47). The manual snow accumulation observations were obtained by request from the NWS MQT Weather Forecast Office. NEXRAD radar data were downloaded from the National Centers for Environmental Information (https://doi.org/10.7289/V5W9574V). An archive of all PIP observations (King & Pettersen, 2023) is available from https://doi.org/10.7302/37yx-9q53. An archive of the MRR observations at Marquette (Richter & Pettersen, 2024) is available at: https://doi.org/10.7302/4qdq-8184.

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